Mathematics - Graduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. How many ways are there to arrange 6 pairs of parentheses such that they are balanced?
   (a) 42
   (b) 144
   (c) 156
   (d) 132
   (e) 96

2. For any poitie integer n, let $\langle n\rangle$ denote the closest integer to $\sqrt{n}$. Evaluate $\sum_{n=1}^{\infty} \frac{2^{\langle n\rangle}+2^{-\langle n\rangle}}{2^n}$.
   (a) 7.0
   (b) 2.0
   (c) 4.5
   (d) 1.0
   (e) 5.0

3. If x and y are directly proportional and x = 3 when y = 8, what is the value of x when y = 13?
   (a) 2.0
   (b) 3.25
   (c) 34.667
   (d) 7.5
   (e) 6.125

4. What is the number of equivalent parameter settings due to interchange symmetries in a mixture model with 10 components?
   (a) 12000000
   (b) 5000000
   (c) 2000000
   (d) 40320
   (e) 7257600

5. What is the ones digit of $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9$?
   (a) 9

(b) 8

(c) 0

(d) 1

(e) 6

True / False (10 marks)

Answer True or False

6. True or False: The eigenvalues of the matrix A = [[5, 4], [1, 2]] are 2 and 6.

7. True or False: The cost of the used car Manuel bought can be expressed as $6,490.

8. True or False: Molly's final cost after a $25 haircut and a 15$

8. True or False: The decimal $0.1\overline{7}$ is equivalent to the fraction $\frac{7}{33}$.

9. True or False: The number 300 is a powerful integer since it contains prime factors whose squares are factors of 300.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In the geoboard shown, the points are evenly spaced vertically and horizontally. Segment AB is drawn using two points, as shown. Point C is to be chosen from the remaining 23 points. How many of these 23 points will result in triangle ABC being isosceles? [asy] draw((0,0)–(0,6)–(6,6)–(6,0)–cycle,linewidth(1)); for(int i=1;i<6;++i) {for(int j=1;j<6;++j) {dot((i,j));} } draw((2,2)–(4,2),linewidth(1)); label("A",(2,2),SW); label("B",(4,2),SE); [/asy]

12. Given a rectangle with length $3x$ inches and width $x+5$ inches, derive the value of x for which the area equals the perimeter. Provide a detailed explanation of your approach.

End of Paper